

Sentinels of Change

2024 Light Trap Network Report

Prepared for the Sentinels of Change 2024 Community Partners



Light trap training in Daajing Giids, Haida Gwaii - Photo by Shaun MacNeill

Report prepared by Heather Earle and Lauren Krzus | Hakai Institute

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Project Information

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PNW Crab Research Group (Puget Sound network): <https://www.pnwcrab.com/larval-crab-study>

Key Points

- 2024 was the third year of the Sentinels of Change light trap network, a network of community partners using light traps to monitor Dungeness crab megalopae along the coast of British Columbia.
- Numerous communities and organizations participated, with 28 light trap stations sampling consistently through the season, 31 partner organizations, and over 250 participants taking part in 2024.
- Partners operated and checked light traps from April 15th to September 1st, 2024 with the exception of some partners who continued monitoring light traps into September as they were still catching Dungeness megalopae.
- 181,319 Dungeness were recorded across the network during the 2024 season, with peak larval arrival occurring during the month of July.
- Catch was highest at traps in the southern Gulf Islands and lower mainland.
- The largest average megalopae carapace widths were from larvae that arrived in April and May with sizes gradually declining throughout the season.
- In June 2024, Hakai staff at the Hakai Marna Lab on Quadra Island conducted a heat stress experiment with juvenile Dungeness crab in mesocosms. Analyses from this investigation are still underway.
- After a successful pilot season in 2023, six light trap stations collaborated with DFO's Aquatic Invasive Species Program to collect and filter water samples from light traps to look for invasive European Green Crab (*Carcinus maenas*) larvae using DNA analysis throughout the 2024 season.
- Light trap coordinators and Hakai project leads debriefed at the Quadra Centre for Coastal Dialogue in October 2024.

*304 data records are excluded from this final tally due to issues with equipment or protocols. More on this below.

Overview

The Sentinels of Change light trap network is an initiative tracking the arrival and abundance of the last larval stage of Dungeness crab (*Metacarcinus magister*), the megalopal stage. Since 2022, when the network was formed as an expansion to the [Pacific Northwest Crab Research Group's](#) existing light trap monitoring program in Puget Sound (PCRG; Buckner et al., 2022), this unique transboundary project has connected scientists, resource managers, and communities spanning the coastline from our northern sites in Haida Gwaii to the bottom of Puget Sound in Washington.

In 2024, the third year of data collection, we continued our efforts to collect high quality data while also expanding the Sentinels of Change network beyond the Salish Sea to other areas of the BC coast. We partnered with new organizations and communities and saw a high return of the partners that took part in previous years. We also piloted additional research that leverages the light trap network to investigate other important questions about threats to Dungeness populations including the impacts of climate change and invasive species. The outcomes of the 2024 data collection efforts are detailed in this report.

Why This Research Matters

This work provides insights into:

- Larval dispersal and population dynamics - Allowing us to investigate how Dungeness crab larvae disperse and how populations fluctuate over time.
- Climate change impacts - Identifying how changing ocean conditions affect various life stages of Dungeness crab.
- Anticipating adult abundance - Past research shows that megalopal abundance correlates with commercial crab landings four years later in some regions (Shanks et al., 2010).

Dungeness crab hold enormous ecological and cultural importance throughout their range and are among the highest value single-species fisheries in both the US and Canada (Norton et al., 2023). They are an important food item for many Indigenous communities and have been for millennia (Ban et al., 2017). The data and connections produced by this transboundary work are critical to fill information gaps and build tools to assist managers in maintaining healthy stocks in the face of climate change and the increasing pressure on crustacean fisheries worldwide (Boenish et al., 2022).

Sites and Methods

From April 15th to September 1st, 2024, community partners deployed and monitored light traps at 28 sites across the coast (Figure 1). This time period covers the known larval delivery season of Dungeness megalopae and is consistent with PCRG protocols. Traps were typically checked (emptied) every two days and each time Dungeness megalopae and first juvenile instars were caught, they were counted and up to 30 megalopae were photographed to be measured using an image analysis software. Other species were identified and counted within the abilities and capacity of each group at each station.

This work takes place across the traditional territories of numerous First Nations who have stewarded their lands and waters since time immemorial.

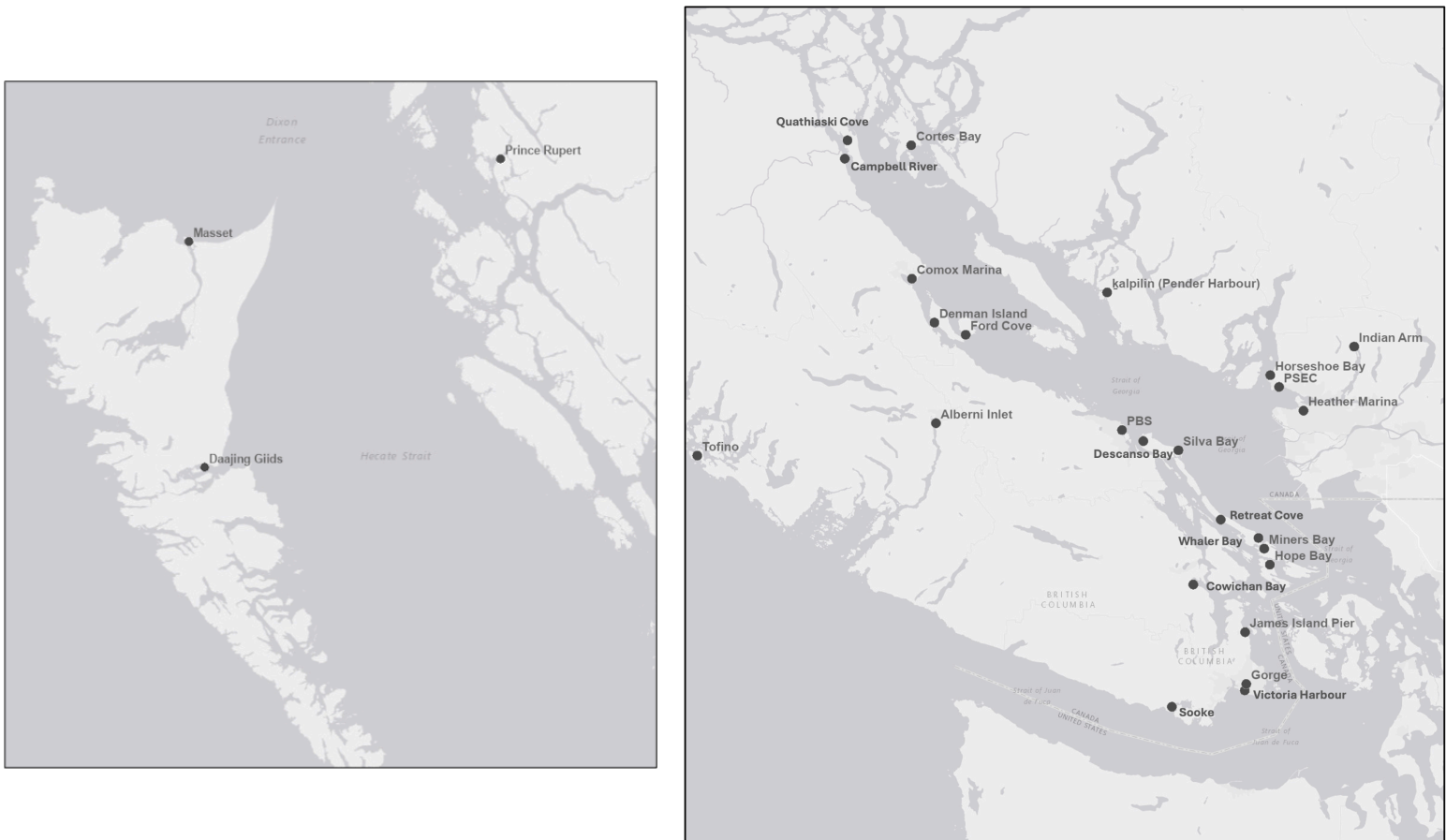


Figure 1. 2024 Sentinels light trap sites along Haida Gwaii and the North coast of BC (left) and along the west coast of Vancouver Island and the Salish Sea (right).

Partners

In 2024, the light trap network had more than 250 participants from a wide variety of organizations and communities, including:

Wigwam Inn, Victoria International Marina, T'sou-ke Marine Team, Tsleil-Waututh First Nation, Tsawout First Nation, Shaw Centre for the Salish Sea, Sewell's Marina, Quadra Island Harbour Authority, Quadra Island Community, Pacific Science Enterprise Centre, Pacific Biological Station, Page's Resort & Marina, Pender Island Conservancy, Pender Harbour Ocean Discovery Station, Ocean Wise, Ministry of Water, Lands, and Resource Stewardship, Mayne Island Conservancy, Jamie's Whaling Station, Institute for Multidisciplinary Ecological Research in the Salish Sea, Hornby Island Natural History Centre, Hornby Island Diving, Ha'oom Fisheries Society, Gorge Waterway Action Society, Galiano Conservancy Association, Gabriola Power Squadron, Gabriola Island Community, Friends of Cortes Island Society, Fisheries and Oceans Canada in Prince Rupert and Port Alberni, Fairway Gorge Paddling Club, False Creek Friends Society, Discovery Passage Aquarium, Denman Island Community, Cowichan Bay Maritime Centre, Cowichan Estuary Restoration & Conservation Association, Council of the Haida Nation, Conservancy Hornby Island, Comox Valley Marina, Centre for Indigenous Fisheries, Átl'ka7sem/Howe Sound Marine Stewardship Initiative, Association for Denman Island Marine Stewards.



Left; three members of the light trap monitoring team on Quadra Island (photo submitted on Device Magic). Right; light trap training in Masset in March 2024 (photo by Shaun MacNeil)

Key Observations

- Larval Dungeness counts in 2024 were similar to observations in 2023. A total of 181,319 Dungeness was recorded (megalopae and first juvenile instars) in 2024*, compared to 229,166 in 2023.
- Peak larval arrival across the network occurred during the month of July.
- In 2024, the majority of larvae were caught in traps in the Southern Gulf Islands (Whaler Bay, Galiano Island and Miners Bay, Mayne Island), as well as in the lower mainland (Horseshoe Bay and PSEC in West Vancouver).
- Larvae were first caught on Vancouver Island in Tofino and in Victoria on April 17th. The last were caught on the Sunshine Coast on September 19th.
- Temperatures tended to be higher at more northern sites and slightly cooler in the more southern portions of the network.
- Sites with high mortality in 2023 were supplied with modified traps and 'heatwave protocols' were introduced for when high catch coincided with high sea surface temperatures. These new measures may have contributed to the 20% reduction in mortality in 2024.

*304 data records are excluded from this final tally due to issues with equipment or protocols. More on this below.

Data Access

Our Dungeness count, size, and temperature data collected through this network are now publicly available on the [Hakai Data Catalogue](#). This dataset will be updated annually after the conclusion of each data collection season. If you have any questions about the dataset, please reach out.

Timing of Megalopae Arrival

Patterns of arrival timing were similar in 2024 to the previous two seasons of monitoring. In 2024, megalopae were caught first on the outer coast of Vancouver Island, in the Strait of Juan de Fuca, and in the Southern Gulf Islands. With the addition of new locations in Tofino and Victoria in 2024, megalopae were caught earlier than in previous years (April 17th compared to early May in 2022/2023) (Table 1, Figure 2). These new sites on the outer and southern coast of Vancouver Island are well situated to catch early arriving megalopae delivered via the California Current along the west coast of Vancouver island before being pushed into the Strait of Juan de Fuca. One of the most northern sites in the network, Masset in Haida Gwaii, also caught megalopae early, in May.

Megalopae arrival timing was similar to previous years for sites in the lower mainland, and along the east coast of Vancouver Island with first catches recorded in early to mid June in all years. We also saw megalopae appearing earlier on the Sunshine Coast and later in some Southern Gulf Island sites than in previous years. As in 2023, the last site to catch Dungeness megalopae was Indian Arm, at the end of the Burrard Inlet on June 26th.

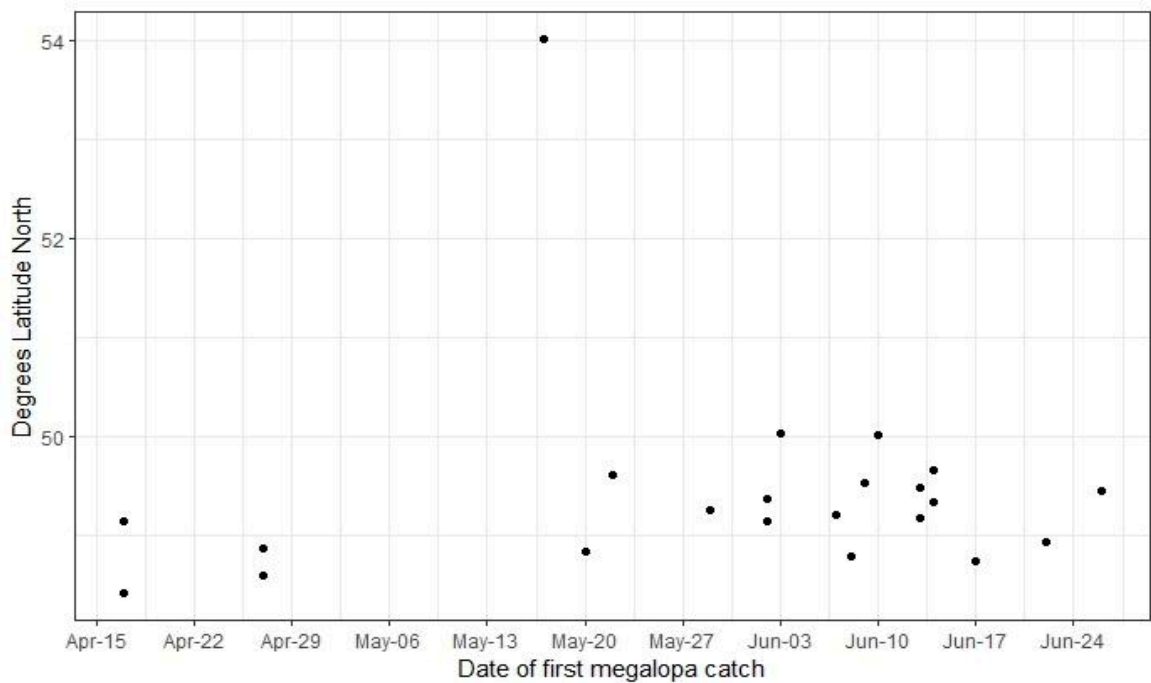


Figure 2. Comparison of first megalopae catch dates at each site latitude. Note that sites that joined the network late in 2024 (after April 15th) and sites that did not catch Dungeness crabs in 2024 are removed from this graph to keep comparisons consistent. This includes: Gorge Waterway, Sooke, Cortes Island, Daajing Giids, Port Alberni, and Prince Rupert.

Table 1. Total catch and first catch summarized by light trap site and year. Note that some sites had equipment malfunctions or joined the network late and therefore may have missed earlier Dungeness arrival as well as later season catch (noted with asterisks). Grey cells indicate sites that did not have a trap in the water in a given year.

Light Trap Site	First Catch 2022	First Catch 2023	First Catch 2024	Total 2022	Total 2023	Total 2024
Alberni Inlet		NA	NA		0	0
Bamfield		May 18*			14	
Baynes Sound		June 6	June 9		535	56
Boot Cove	July 1*			438		
Campbell River		June 5	June 10		1,227	152
Comox Valley			June 14			11
Cortes Bay	June 12	June 18	NA	75	8	0
Cowichan Bay	June 27	June 1	June 17	421	4,936	190
Daajing Giids			NA			0
Descanso Bay	June 14	June 4	June 13	274	3,445	12
False Bay	July 9	May 31		37	2,061	
Ford Cove	July 4	June 1	June 13	396	329	61
Gibsons Harbour		June 10			568	
Gorge Waterway			May 31*			2
Heather Marina	June 22	June 8	May 29	1,207	329	852
Heriot Bay	June 23	June 16		11	175	
Hope Bay	June 4*	May 11	June 8	5,015	2,496	6,618
Horseshoe Bay	June 20	June 8	June 2	4,081	83,627	21,099
Indian Arm		June 26	June 26		78,893	8,673
Institute of Ocean Sciences	May 25			83		
James Island Pier		May 4	April 27		1,712	7,072
Lyll Harbour		May 10			205	
Masset			May 17			7
Miners Bay	May 11	June 2	May 20	4,550	8,343	39,367

Pacific Biological Station	June 20	June 2	June 7	105	105	31
Pacific Science Enterprise Centre	June 20	June 5	June 8	3,351	34,728	44,222
kalpilin (Pender Harbour)	June 16	June 14	May 22	110	584	157
Porpoise Bay		June 14			547	
Powell River		August 22*			1	
Prince Rupert		NA	NA*		0	0
Quathiaski Cove			June 3			3,393
Retreat Cove	May 9	May 13	June 20	52	63	9,607
Silva Bay	June 7	May 30	June 2	1,754	2,600	4,522
Sooke Harbour		May 12	May 29*		21	122
Surge Narrows	June 10	NA*		20	0	
Tofino			April 17			11,700
Victoria			April 17			89
Whaler Bay	May 12	May 5	April 27	587	334	23,918
Winter Cove	May 9			163		

Megalopae Abundance

Across the network, we recorded a total of 181,319 Dungeness crabs over the 2024 season. This number includes both megalopae and first juvenile instars (megalopae often moult into juveniles inside the trap). The majority of the catch occurred between mid-June and mid July which was slightly earlier than 2023 where we saw the most Dungeness between mid June and early August. Catch diminished through August but, unlike previous years, megalopae continued to show up in light traps at James Island Pier and Pender Harbour into September. (Figures 3 and 4).

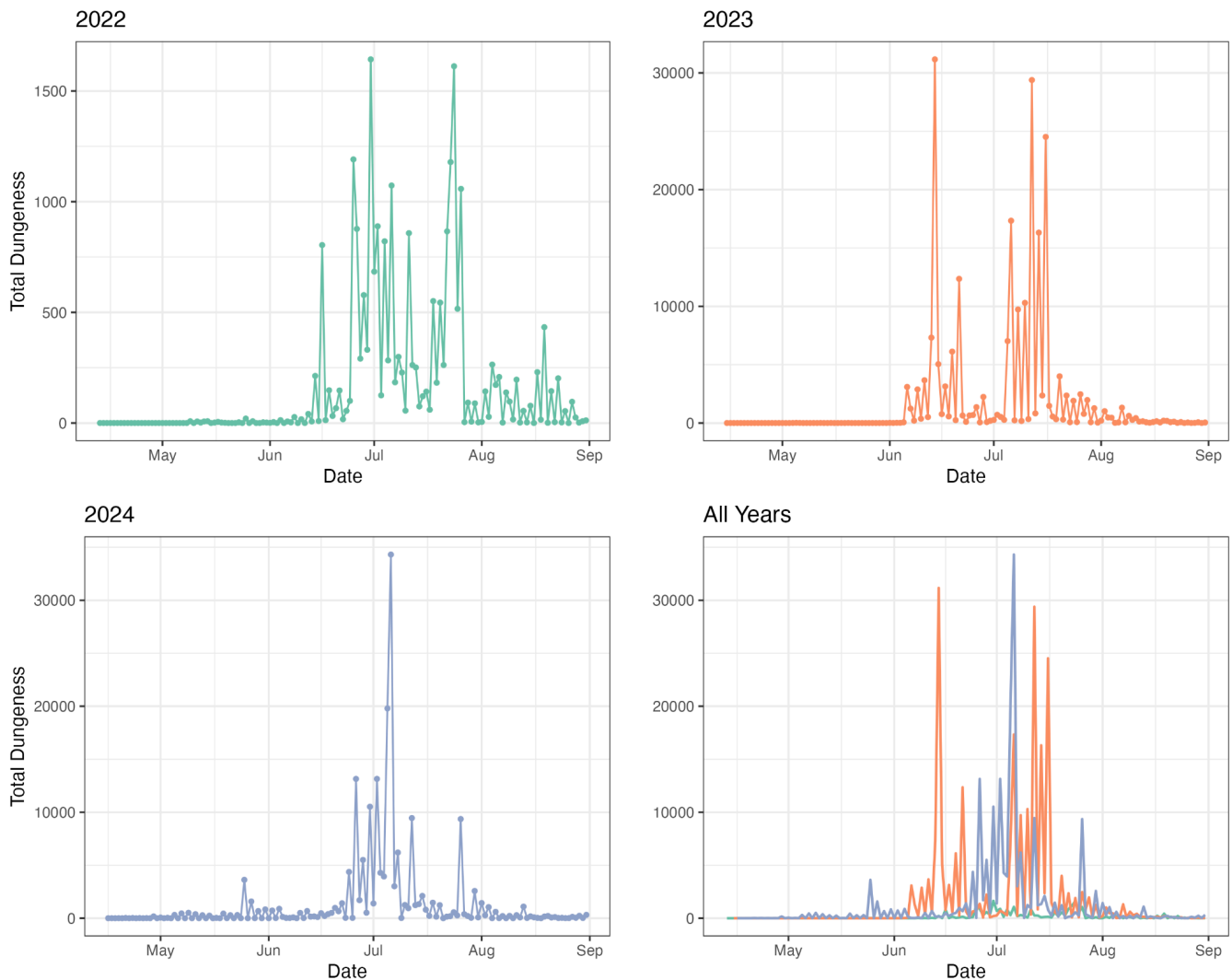


Figure 3. Total Dungeness (megalopae and first juvenile instars) caught across all sites by year. Note the differences in the vertical axes between years.

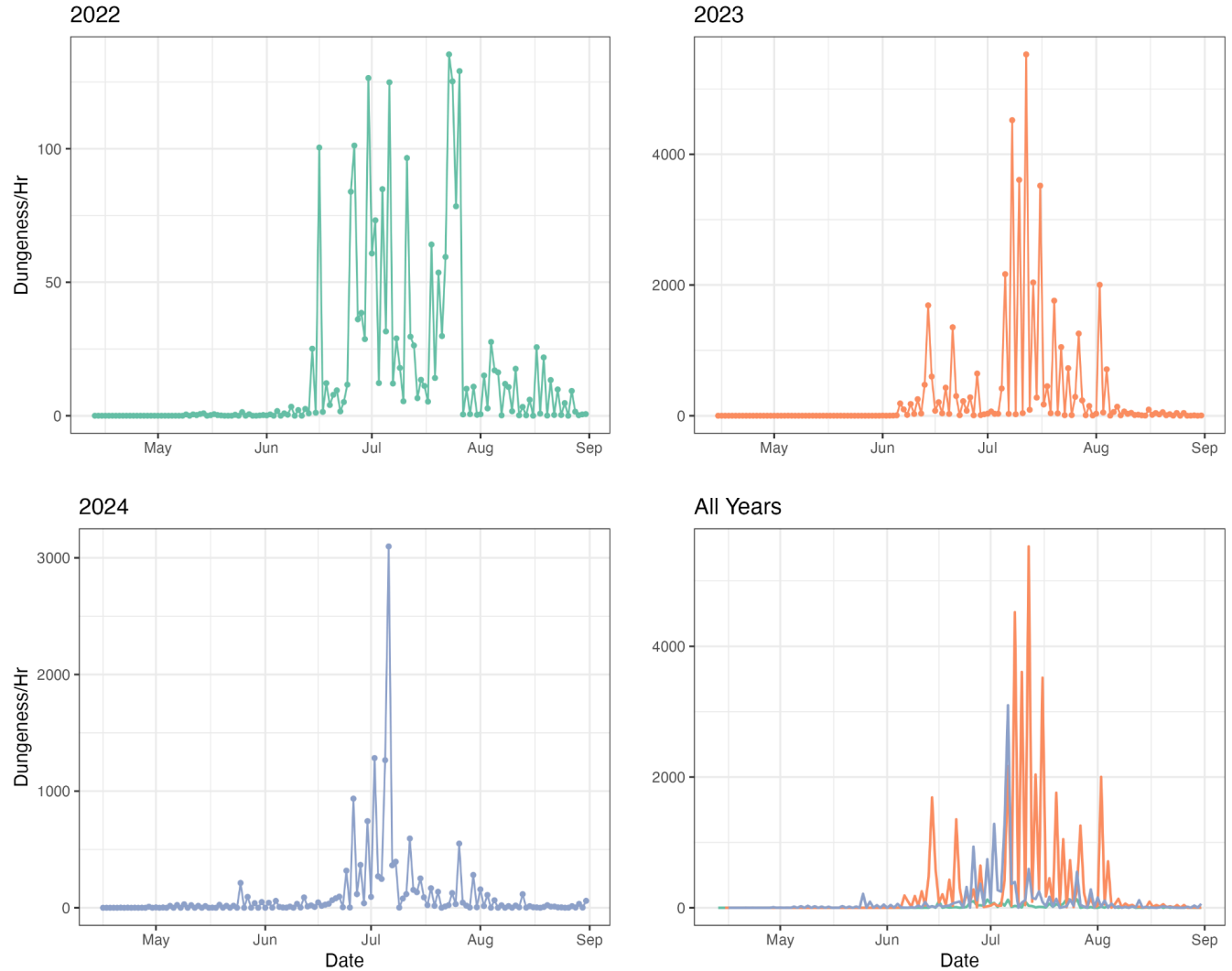


Figure 4. Catch per unit effort (CPUE) (megalopae and instars per hour) across all sites by year. CPUE was calculated as the number of crabs caught per hour when a trap is “fishing” (lights on) and thus accounts for variation in fishing effort across sites. Note that effort changed across the season as we adjusted the timers with sunset and sunrise through the season. Note the different vertical axes between years.

Megalopae Catch by Site

In 2024, many sites across the network experienced smaller catches compared to 2023 including sites in the central Strait of Georgia (Nanaimo, Galiano, Hornby Island and Denman), the Sunshine Coast, and Cowichan Bay. Sites in the lower mainland (Horseshoe Bay, Heather Marina, Indian Arm, and PSEC) caught the highest numbers of Dungeness (74,846 combined) (Figures 5, 6 and 7). The lower mainland sites accounted for 41% of the total catch in 2024 compared to 86% of the total catch in 2023. In contrast, the Southern Gulf Island region (Miners Bay, Retreat Cove and Whaler Bay) had their biggest year yet collectively catching 72,892 Dungeness and making up 40% of the entire catch of the season compared to 4% in 2023. With the addition of new sites in Tofino and Victoria there were large catches earlier in the season with over 11,000 megalopae caught before July. In 2024, the light trap on Quadra Island was moved from Heriot Bay, on the east coast of the island, to Quathiaski Cove, on the west and caught almost 20 times more Dungeness than in 2022 and 2023 combined. Across the Discovery Channel, in Campbell River, only 152 Dungeness were caught in 2024 compared to 1,227 in 2023.

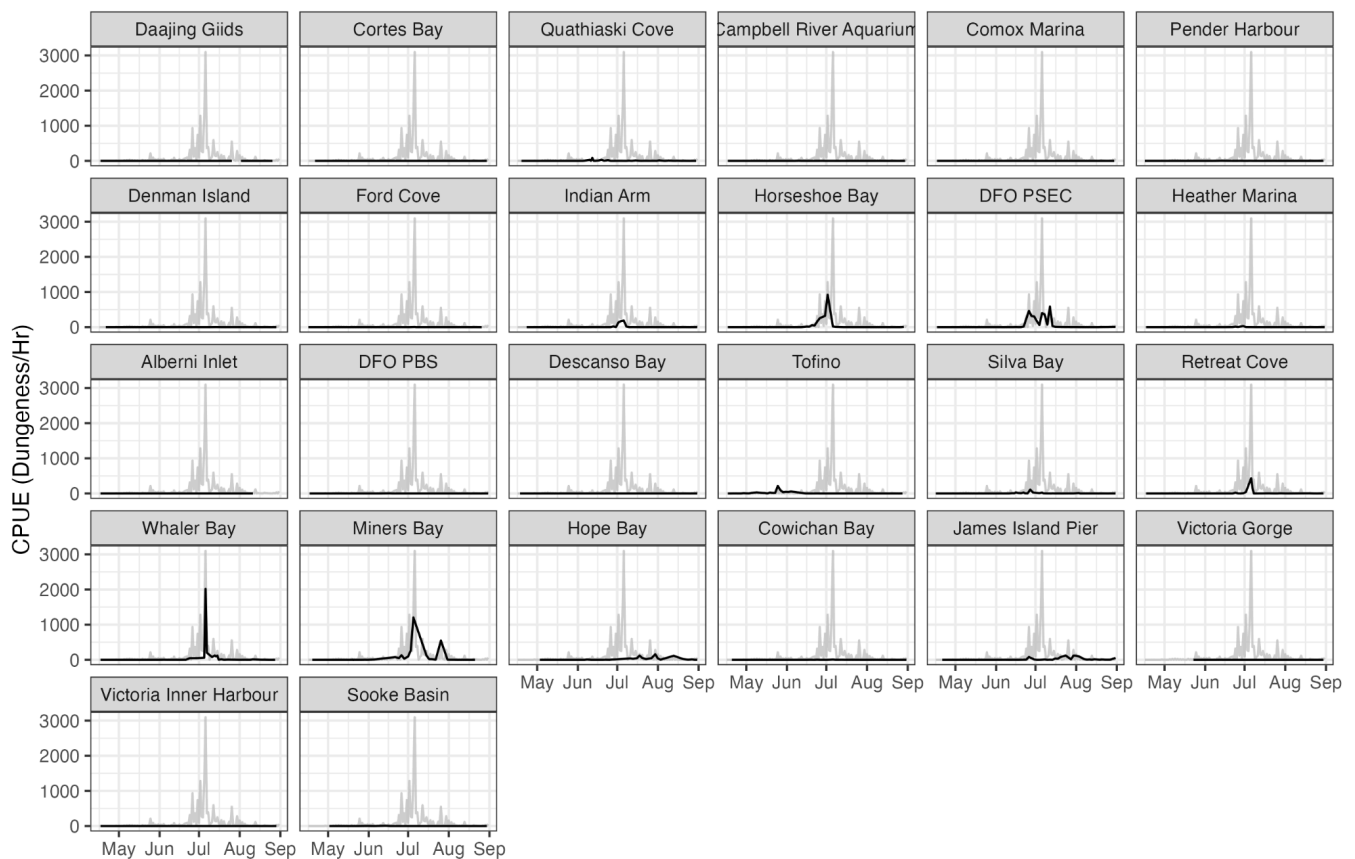


Figure 5. Dungeness CPUE (crabs per hour) shown at each site in the network (black lines). Sites are ordered by latitude with most northern sites at the top and most southern at the bottom. The grey lines are the same in each panel and show the total CPUE across all sites. Sites with an incomplete data set have been excluded, this includes: Masset and Prince Rupert.

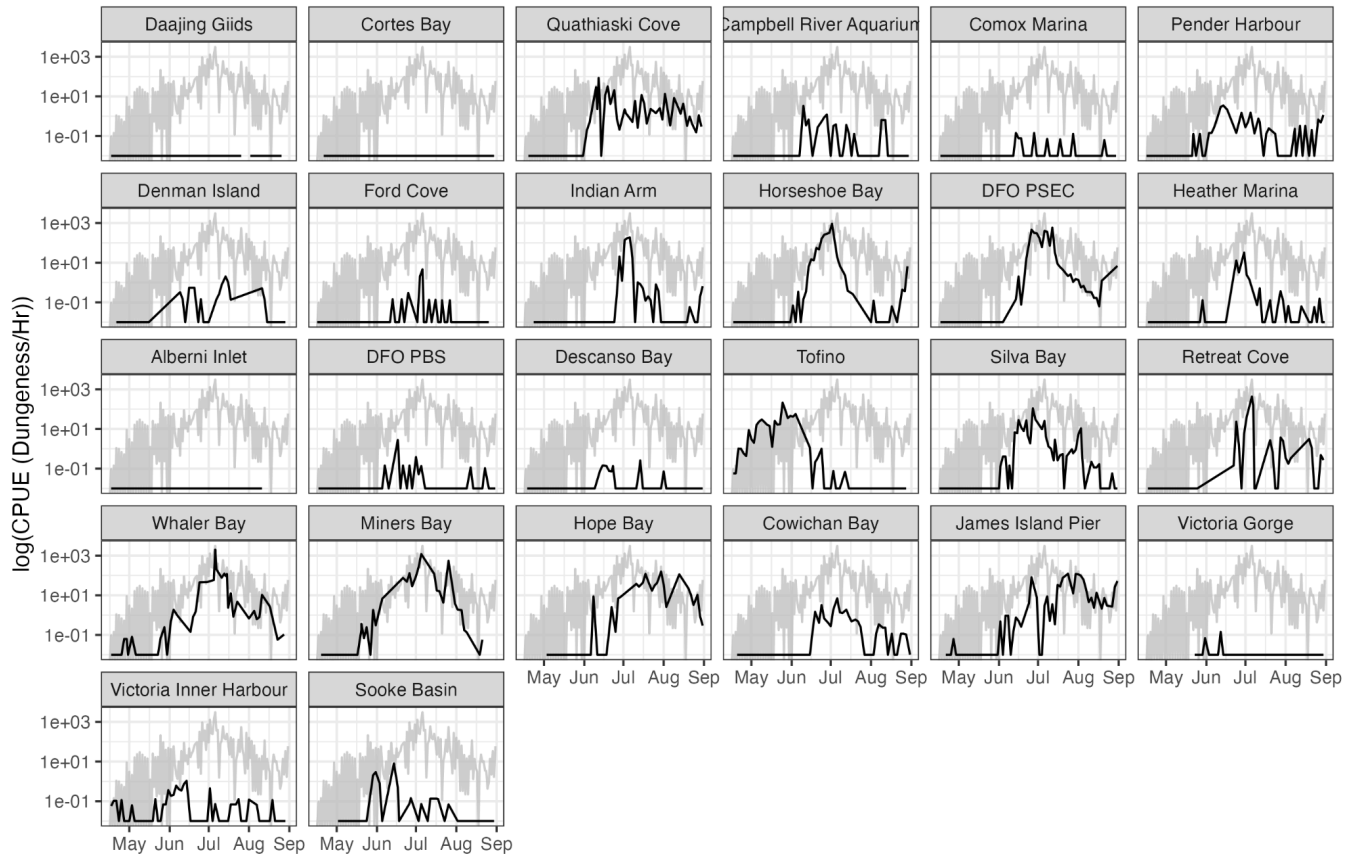


Figure 6. Dungeness catch per unit effort (crabs per hour) shown at each site in the network (black lines). In this figure, the data are displayed on a logarithmic scale so that trends across sites with lower catch can be seen. Sites are ordered by latitude with most northern sites at the top and most southern at the bottom. Sites with incomplete datasets have been excluded, this includes: Masset and Prince Rupert.



Figure 7. Dungeness catch per unit effort (megalopae and instars per hour) averaged by month (mean CPUE) shown at each site in the network for 2022, 2023, and 2024. The size of each orange bubble represents the mean CPUE for each site in that month. This map excludes the sites that did not collect more than one week of data in any year, as well as the Prince Rupert site which caught no Dungeness in 2023, and northern sites in 2024 which did not catch Dungeness or only had a short partial season (Daajing Giids, Masset, and Prince Rupert)

Potential influences on larval dispersal

Various drivers operating at different scales are known to influence the arrival and abundance of Dungeness megalopae. Much of the research investigating these drivers has been carried out on the west coast of the United States, which differs oceanographically from the BC coast. However, it is likely that many of the processes influencing timing and abundance further South are also relevant along our coastline. Along the West coast of Oregon and California, Dungeness larvae hatch in winter and are pushed offshore such that by the time they moult into megalopae, they are beyond the continental shelf (Shanks et al., 2010). The survival of these larvae and whether they will grow into adults hinges on whether they can make it back across the shelf to shore to settle in shallower waters as juveniles. Different factors influence where these larvae end up and whether they will grow into adults. Below, we present a summary of these drivers and how they relate to our catch data over the previous three seasons.

Timing of the Spring Transition

The spring transition occurs when winter winds from the south are replaced by spring/summer winds from the north along the Pacific coast of North America. This shift replaces the northward flowing Davidson Current on the continental shelf with the southwards flowing California Current as it moves back onto the shelf and begins the cycle of summer upwelling (Shanks et al., 2010). This shift transports larval Dungeness crab back over the shelf to shallow coastal waters where they can settle. The spring transition can occur as early as March to as late as July and researchers have found that the day of the transition is strongly correlated to the size of returns of Dungeness megalopae in Oregon light traps; when the transition is early, the return is large and vice versa (Roegner et al., 2007, Shanks 2013). This is likely because earlier spring transitions transport larvae back onto the shelf earlier, reducing time in the plankton and the likelihood of mortality (Shanks et al., 2010).

Spring transition timing could impact larval returns across the BC coast differently. An early transition could result in more larvae in the California Current some of which would enter the Salish Sea via the Strait of Juan de Fuca, increasing returns in our inland sea. Further North, an early transition may pull more larvae out of the Alaska Gyre System resulting in lower returns. As this project continues, we will continue to compare the timing and strength of the spring transition with the abundance and timing of megalopae arrival across the coast and determine how this driver influences Dungeness crab recruitment in BC.

Pacific Decadal Oscillation

The Pacific Decadal Oscillation is a basin scale climate index that correlates with the amount of water from the North Pacific Drift that is deflected into either the Gulf of Alaska (increased during positive phase) or the California Current (increased during negative phase) (NOAA 2024). Previous work has shown that in years when more water flows into the California Current, the abundance of returning megalopae is higher in Oregon and California (Shanks et al., 2010), likely because more megalopae are transported southward rather than being advected into the Gulf of Alaska system (Shanks 2013).

We are currently in a negative phase of the PDO with the last shift occurring in late 2019 (NOAA 2024). The implications of this to larval delivery will vary along the BC coast. Larvae are known to be advected into the Salish Sea via the Juan de Fuca Strait (Buckner et al., 2025), therefore a negative phase PDO might increase these returns as more megalopae are pushed south rather than north. At our more northern sites, increased movement of water to the south associated with a negative phase could reduce larval returns.

Internal Waves

Work in coastal Oregon indicates that internal waves are possibly the strongest environmental driver of daily catch rates of Dungeness crab (Rasmuson et al., 2020). Unlike surface waves, internal waves are driven by subsurface changes in density such as the sharp gradient found between fresh and saltwater. These waves can travel many kilometres and transport larvae across the continental shelf and may explain why we often find Dungeness larval delivery occurring in 'pulses' rather than arriving neatly with spring/neap tides. On the outer coast the strength of internal waves can also account for the variation in daily megalopae catch (Rasmuson et al., 2020) and it has been speculated that the stronger flow of the California Current during a negative PDO may create conditions that favour strong internal waves and therefore elevate catch numbers (Shanks 2013). We have yet to determine the relative importance of internal waves as a driver of daily catch rates in BC, but it will be one of many factors we consider as we dive into the data.

Freshwater Inputs

Dungeness megalopae are known to be sensitive to salinity as it impacts their growth and development (Rasmuson 2015) and may avoid low salinity water including freshwater surface layers in areas with high freshwater inputs. This is especially relevant in the Salish Sea during spring and summer when melting snowpack produces the Fraser river freshet which fills the Salish Sea with brackish, turbid, and relatively warm water on ebbing tides. As the tides ebb, the Fraser river plume spills out in all directions with strong pulses westward and through Active Pass and

Porlier Pass (Leblond 2012). In 2024, a relatively weak Fraser river plume may account for higher numbers of megalopae recorded in Southern Gulf island traps as the extent of the freshwater lens was likely diminished relative to previous years. We will continue to monitor the relationship between freshwater inputs and megalopae distribution and abundance across the BC coast. In 2025, we will be deploying salinity loggers at many of our sites across the network to monitor this relationship.

Megalopae Size

All megalopae were measured digitally with image analysis software, ImageJ. When megalopae were present, partners placed up to 30 into a standardized white tray and took a level, overhead photo. These images are imported into ImageJ to measure carapace widths. Although there are sometimes issues with tilt and glare with the photos, this approach has been effective in reducing monitoring time and effort for partners and improving accuracy of measurements.

As in 2023, the largest carapace widths were measured from megalopae caught in the months of May and June with sizes declining through the remainder of the season (Figure 8). The largest megalopa measured 5.25mm (July 2nd at Indian Arm) and the smallest 1.96mm (May 24th Miners Bay). In 2023 the largest and smallest megalopae, 4.55mm and 1.71mm respectively, were both caught in mid June.

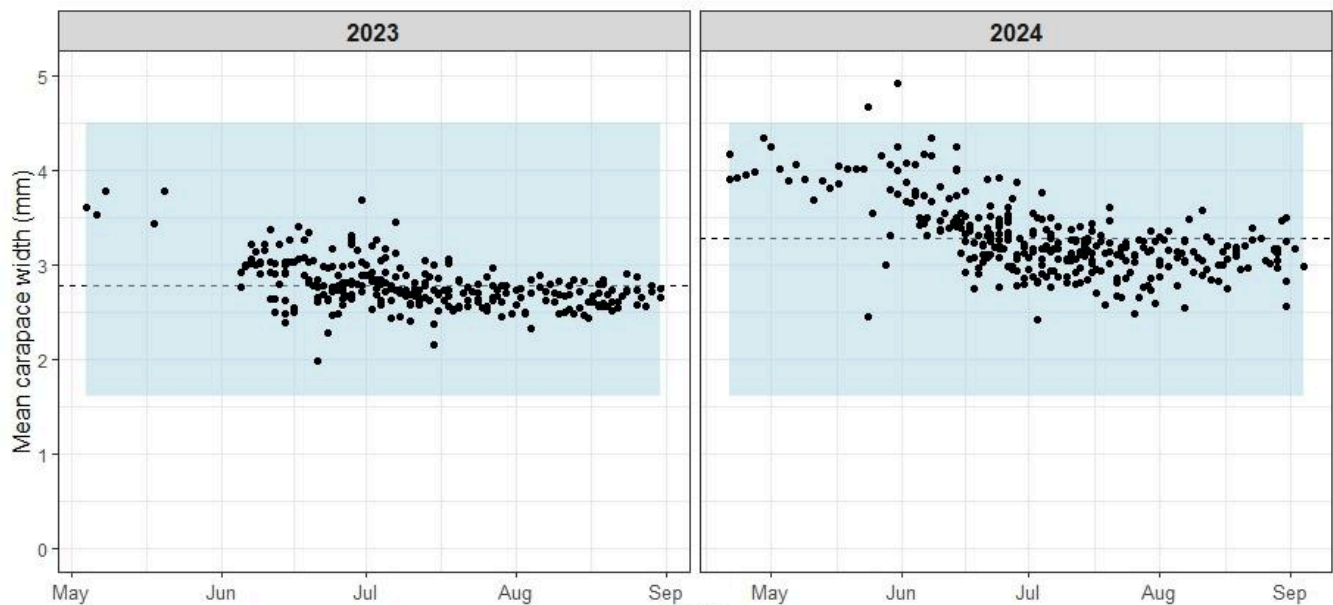


Figure 8. Carapace widths of megalopae measured from all sites across the 2023 and 2024 seasons. Each dot is an average of all megalopae measured on each date at each site. The dotted lines show the average width through the season across all observations, and the shaded rectangle shows the typical range of Dungeness carapace widths from published literature (Debrosse et al., 1990).

In Figure 9, we can see megalopae carapace widths change through the season at each site. The violin plots show the monthly size distribution of measured megalopae. Note that many sites did not catch megalopae in all months and in some months, there were very few megalopae caught resulting in small sample sizes. Photo quality tended to vary and so not all megalopae that were photographed could be measured.

At some sites (Heather Marina, Indian Arm, Masset, Tofino, Retreat Cove and Whaler Bay), megalopae exhibited a gradual decline in carapace width through the season (Figure 9). This aligns with trends seen in 2023 and the hypothesis that an early cohort of larger megalopae come in from the outer coast through the Juan de Fuca Strait and a late cohort of smaller megalopae originate from the Salish Sea appearing in traps later in the season (Dinnel et al., 1993). Overall, megalopae caught in 2024 were larger in each month than they were in 2023, with the largest average megalopae occurring in April 2024. (Figure 10). Interestingly, Cowichan Bay, Denman Island (Baynes Sound), Miners Bay, and Victoria saw carapace widths increase in August, while the other sites remained consistent over the 4 months of monitoring (Figure 9).

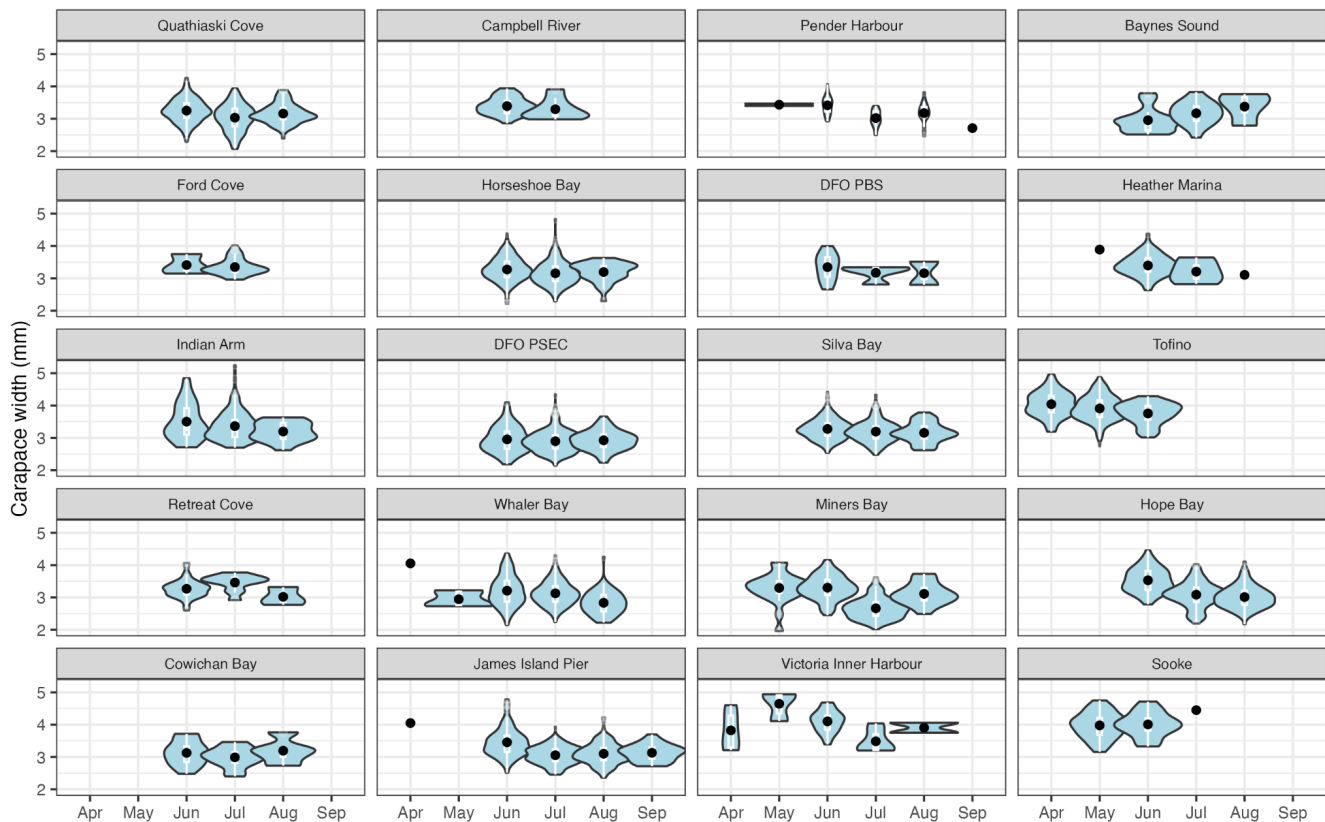


Figure 9. Distribution of carapace widths by month and site in 2024. Note that sites with fewer than 15 megalopae across the entire 2024 season were excluded from this graph. The black dots represent the median carapace width in each month and the blue shaded areas show the distribution of carapace widths. The wider sections of the violins indicate sizes that occurred more frequently than the narrow sections. Sites are ordered by latitude with most northern sites at the top.

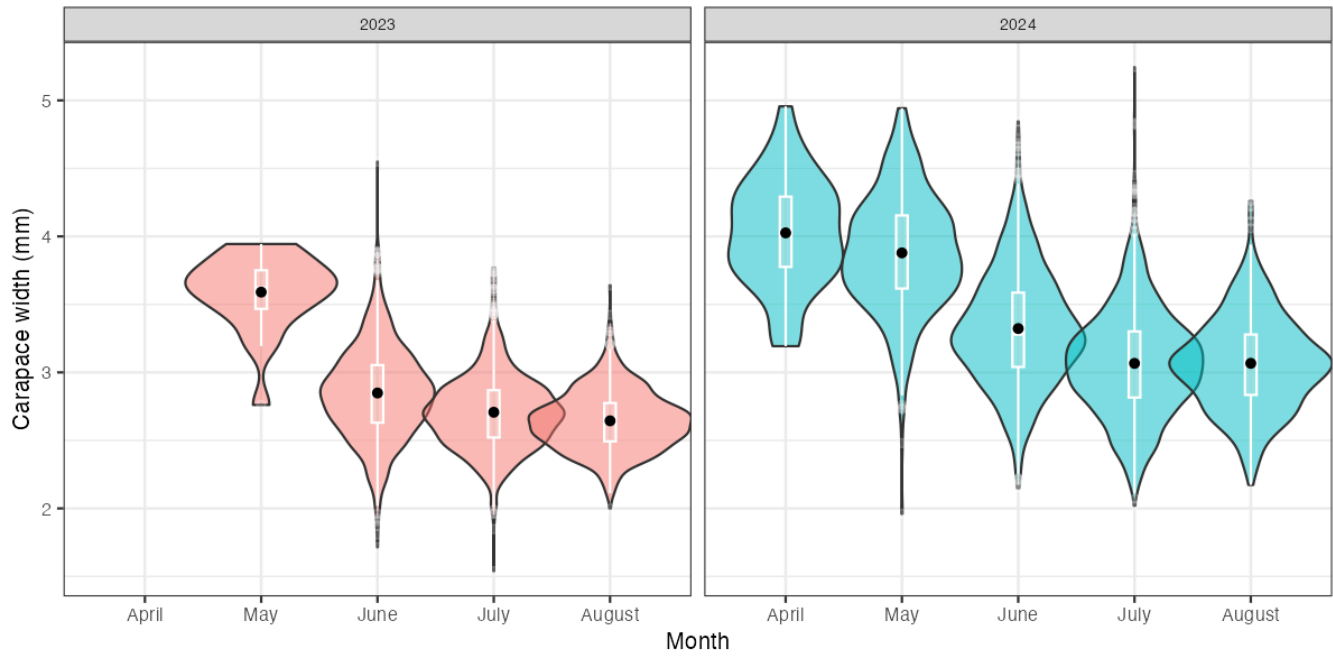


Figure 10. Distribution of carapace width measurements by month from 2023 and 2024. The black dots represent the median carapace width in each month and the coloured shaded areas show the distribution of carapace widths. The wider sections of the violins indicate sizes that occurred more frequently than the narrow sections.

Sea Surface Temperature

We use Onset HOBO TidbiT MX2203 temperature data loggers to record sea surface temperature at each light trap site. Loggers are suspended 0.5m below the surface from floating docks adjacent to light traps and are programmed to record the water temperature every 10 minutes for the entirety of the light trap season. Figure 10 shows temperature trends by site across the season, organized by latitude (northern sites on top and southern sites on the bottom).

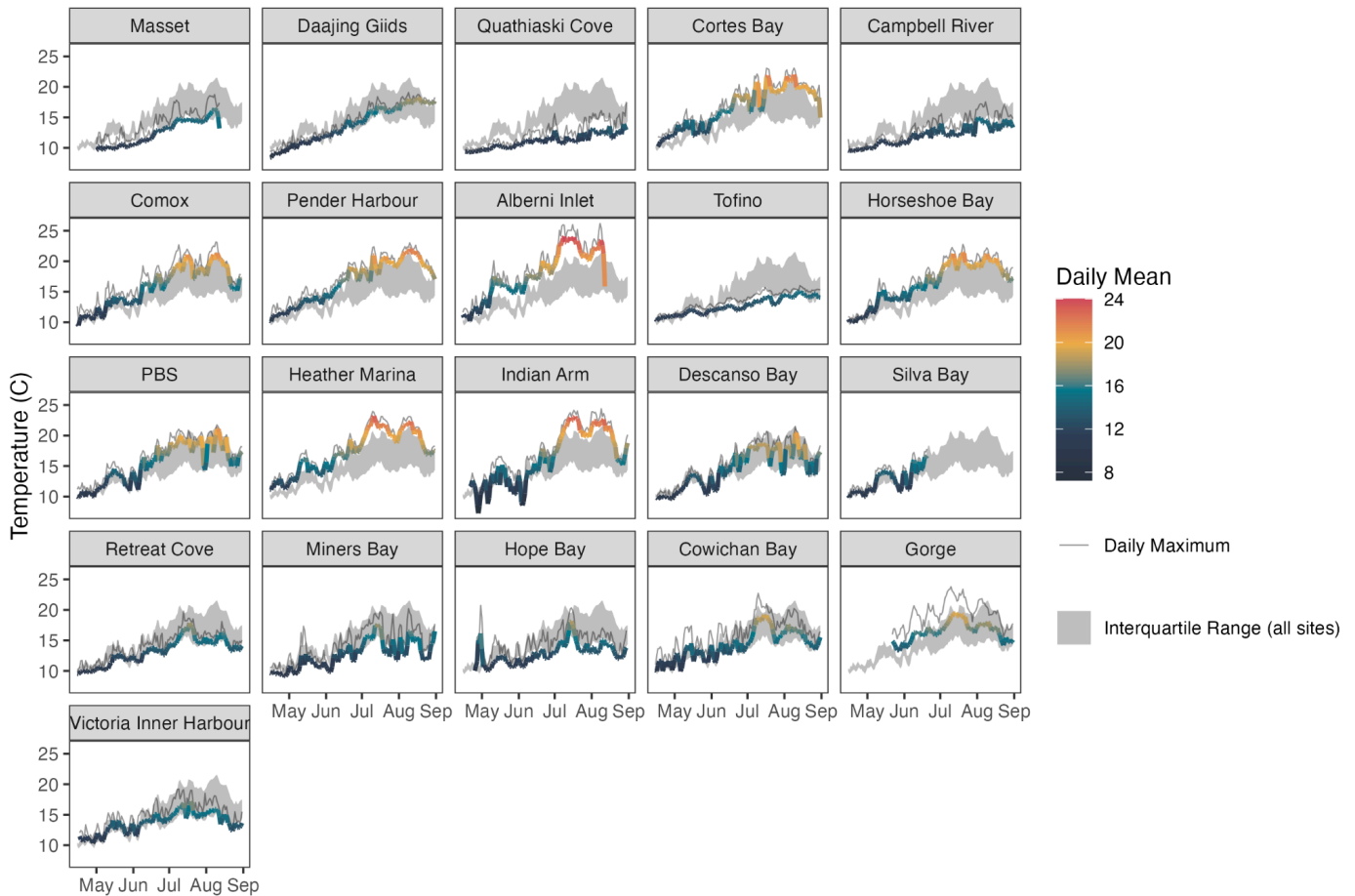


Figure 11. Sea surface temperature recorded at each light trap site from April 15th - September 1st, 2024, descending by latitude from top to bottom. In each panel, coloured lines depict the mean daily temperature at that site, the thin dark grey line indicates the daily maximum at that site, while the grey shaded area represents the interquartile range (middle 50% of the dataset) of all sites. Note that data for some sites are missing as temperature loggers were lost at several sites in 2024.

Temperature and Peak Larval Arrival

Figure 12 shows CPUE through the season at each site coloured by average sea surface temperature on that date. Here, we see that some sites experienced peak larval delivery when sea surface temperatures were near their warmest of the 2024 season including Descanso Bay, Heather Marina, Horseshoe Bay, Indian Arm, and Pender Harbour.

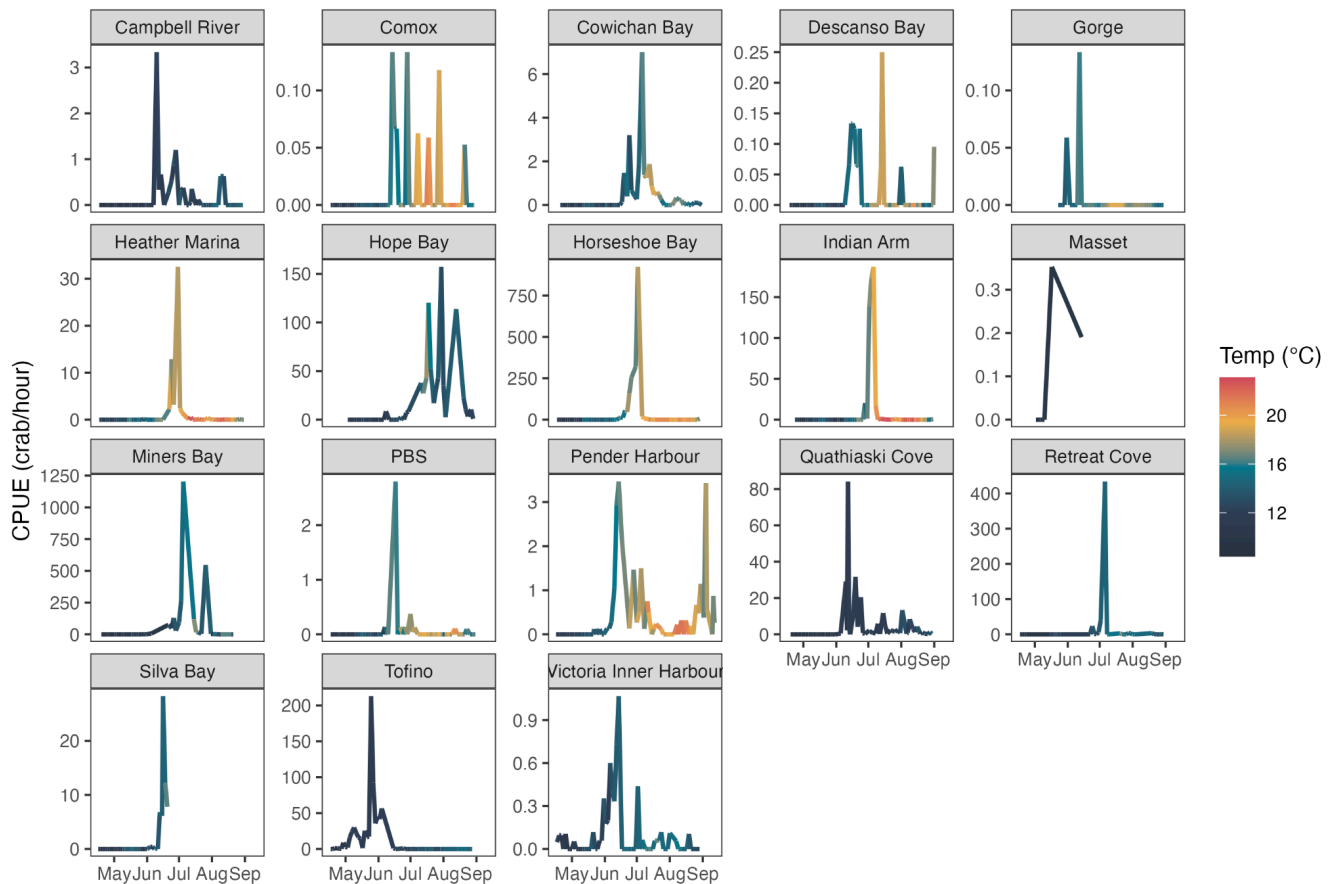


Figure 12. CPUE (crab/hr) at each light trap site coloured by average sea surface temperature, descending by latitude from top to bottom. Note that the y axes differ across plots as CPUE varies considerably across sites. Some sites are not included in this plot if they did not catch Dungeness in 2024 or if temperature data were not yet retrieved at the time this report was released.

Challenges and Improvements

Data Quality

To ensure the quality of our dataset from year to year and that it is comparable to data collected in our partner network in Washington, we review each entry submitted through Device Magic during and after the data collection season. Often, we catch small errors that we can easily address by following up with site coordinators (hence the many emails from us throughout the season!). Sometimes we have to exclude a data entry entirely for reasons such as equipment failure or significant deviations from the trap checking protocols.

In 2024, we excluded a relatively large number of data records—304—from our final counts. Unfortunately, some of these records included the largest catches of the season due to challenges with subsampling. In total, excluded data records led to a reduction in 176,626 Dungeness from our total count in 2024, largely from traps in the Southern Gulf Islands and lower mainland. It's important to note that this may not represent the actual amount of larvae that were caught on those occasions and excluded from the dataset, as we cannot estimate those numbers. Nonetheless, we can still conclude that it is a significant reduction in our totals for 2024.

We will continue to work closely with partners to ensure we are producing quality data and that all protocols are well understood and followed. In particular, it seems that subsampling can be challenging for some groups as it's carried out infrequently through the season and so samplers may be unfamiliar with the protocol. As training gets underway for 2025, we will revisit these protocols with participants and remind them that they are always available to revisit in the laminated protocols provided to each site as well as in video format in the Sentinels google drive.

Equipment Changes

In 2024, we made several changes to equipment and protocols to address the challenges with high mortality of Dungeness crab experienced during elevated sea surface temperatures in the 2023 monitoring season. This included modifying traps at sites with high mortality by adding extra vents to improve water circulation, the creation of a 'heatwave protocol', and supplying sites with a timed aerator to oxygenate water in the traps. We saw a 27% reduction in Dungeness mortality in 2024. We are continuing to investigate other trap and protocol modifications to reduce mortality as we anticipate many more hot summers to come. Other improvements made in 2024 include an updated protocol video, the addition of a zinc anodes to weights to prevent rusting and loss, aluminum foil was added to dowel to prevent light burn outs, and we changed the colour of the measurement trays from black to white to reduce glare.

Community Engagement

During our annual postseason debrief with community coordinators, we discussed how community engagement and volunteer participation can be improved. Next season we will continue to work with coordinators and community members to bring more awareness to the project through advertising, presenting and/or providing materials for local events. Next season we will add prompts to Device Magic recording the number of people and time spent monitoring each trap check as this information can be useful for grant applications to secure funding for consistent monitoring. These data will become available to site coordinators after the 2025 season upon request. Finally, to provide incentive for highschool or graduate student participation in the project, the Sentinels of Change team can provide a certificate or reference letter for students hoping to add light trap monitoring to their resume or contribute to volunteer hours.

We are currently working on other improvements to equipment and protocols for the 2025 season, which are outlined in Table 2. Thank you to all participants that provided feedback and suggestions during the 2024 season!

Table 2. Equipment challenges and identified/implemented solutions.

Problem	Solution
Bucket flooding and organisms getting trapped	Reduce weight, standardize foam volume and seal foam to bucket
Organisms getting caught in small spaces	Seal off problem zones with silicon
Difficulty sampling alone	Provide a trap stand upon request
Timer programs being wiped	Tape wires out of the way and/or find timers with protective case
Cod end draining slowly	Coarser mesh in cod end

Other Species

Light traps attract many other interesting species of which the documentation varies with the capacity and interest of participants at each site. Participants record other species in Device Magic as well as through our iNaturalist community project where participants upload pictures of unknown (or known) species caught in light traps. iNaturalist is an online platform with a crowdsourced species identification system. In 2024, the light trap network contributed 323 observations and identified 86 species. Overall, the project has made 718 observations of 118 species since it was launched in 2022! Thank you to everyone who contributed to the iNaturalist community project!

[Explore the Sentinels of Change iNaturalist page here!](#)



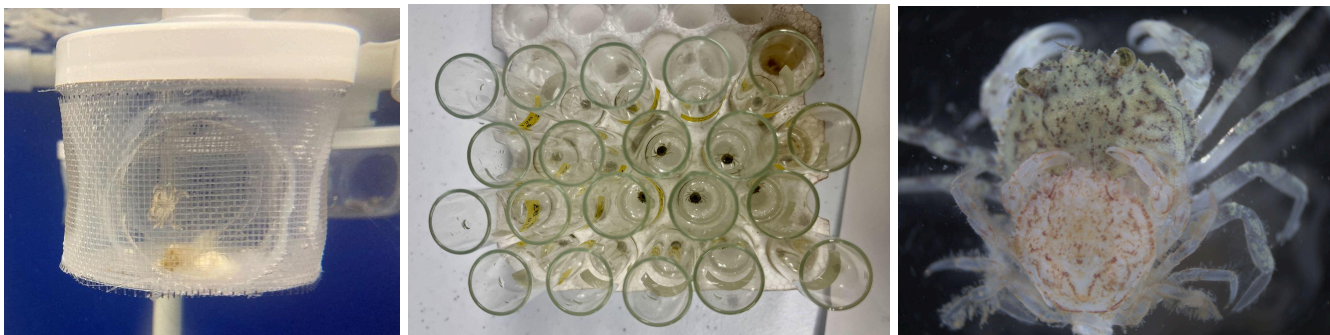
Species caught in light traps, photographed by various participants, and identified on iNaturalist.

Additional Work in 2024

Mesocosm Experiments

After a successful pilot project in 2023 to establish protocols for collecting, transporting, and maintaining Dungeness megalopae in wet lab mesocosms, we conducted a heat-stress experiment with juvenile Dungeness at the Hakai Marna Wet Lab on Quadra Island in 2024. We collected Dungeness crab megalopae from the Quadra Island light trap and held them in the Marna lab until they molted into juvenile instars before being separated into short and long heat wave treatments. For both treatments, we raised the temperature from 15°C to 25°C over 8 hours. Crabs in the short heatwave treatment were held at 25°C for 32 hours, while the long heatwave treatment were held at temperature for 60 hours.

We housed all 360 crabs used in this experiment in custom made 'crabitats' (pictured below) and fed them a diet of mussels harvested from the Hakai Shellfish Raft in Hyacinthe Bay. The wet lab team monitored feeding rates, moulting, and mortality, and took photos and collected tissue samples before and after the heatwave treatments to compare visual and molecular changes over the experiment. We are still processing the data from this experiment and we plan to update the network with results when they are available. In the coming years, the Marna lab will continue to work with Dungeness crab to explore their vulnerabilities to the impacts of climate change and provide insight into how local populations will fare in a changing ocean.



Left: A 'crabitat' with a crab inside; middle: juvenile instars separated into individual test tubes and right: a moulting juvenile instar, photos by Rumer Opie.

Dungeness Pathogen Investigations - Update from PhD Student Allie Roberts-Moore

The 2024 season was really successful for our project. In total, we collected just over 400 megalopae from Horseshoe Bay and Pender Island. We cleaned these crabs off in filtered seawater before storing them in a solution that stabilises RNA in the -80 freezer until we were ready to extract RNA and DNA. Once the RNA and DNA has been extracted from the crabs, we can sequence the RNA to look for sequences that look like known viruses, and test to see if these

potential viruses are interacting with the host immune system. With the DNA, we will be looking to see what types of microorganisms are associated with the larval crabs and how that's related to their viruses. We're excited to go back this year and collect more samples from Horseshoe Bay. Special thanks to Riley Plunkett, Isaak Haberman and the Marine Stewardship Initiative for help with sampling for this project in 2024!



Allie Roberts-Moore collecting Dungeness megalopae to look for viruses.

Dungeness Investigations in Tsawout Territory - PhD student Nicole Jung

Together with the Tsawout First Nation community and Fisheries Department, and under the supervision of Dr. Andrea Reid with the UBC Centre for Indigenous Fisheries, PhD student Nicole Jung investigates Dungeness crab (ÁĆEX) lifecycle and populations with the goal of producing a stewardship protocol for Tsawout's QEN'T Indigenous Protected & Conserved Area. To fill in knowledge gaps around local Dungeness crab populations, she is conducting surveys at different life stages using light traps, modified commercial traps, and previously collected datasets. This work is strengthened by interviews with Tsawout crab harvesters, Elders, and knowledge holders to deepen collective understanding of historical abundance and to ensure the protocol properly reflects the values and cultural practices of Tsawout. This work also involves the development of a youth internship program, creating opportunities for Tsawout youth to learn from community knowledge holders and Tsawout Fisheries staff about Tsawout's deep-time and ongoing relationships with their home-lands and -waters.

European Green Crab Detection

In August and September 2023, partners at 6 sites (Cortes Island, Descanso Bay, Ford Cove, Pender Harbour, Sooke Harbour, and James Island Pier) worked with DFO's Aquatic Invasive Species program to pilot a project to detect invasive European Green Crab (EGC; *Carcinus*

maenas) larvae in light traps. As EGC larvae are too small to identify visually, partners collected and froze water samples from light traps for genetic analysis to determine if EGC larvae are present. Genetic analyses of our 2023 water samples only produced positive detections of EGC DNA in Sooke where there is an established adult EGC population.

In 2024, the same partners collected water samples every 2 weeks throughout the monitoring season. DFO recently shared the results with us which did not provide evidence of established populations at any of the sites except Sooke. They did detect very low levels of EGC DNA at the Gabriola, Saanich Peninsula, and the Sunshine Coast sites. The low levels of DNA detected in the analyses indicate that EGC at these sites are rare. Given the sensitivity of these tests, a positive result is not enough to confirm that adult EGC are present. They do, however, warrant a follow up investigation and this work will continue into 2025.



Juvenile EGC caught in the light trap in Daajing Giids, Haida Gwaii.

Media, Outreach, and Publications

2024 articles, blog posts and talks

- Blue Water Cruising Association: '[Vancouver Hybrid Club Night - Sentinels of Change](#)'
- Fisheries Journal: [Developing a collaborative Dungeness crab larval monitoring network in the Salish Sea to provide long-term, fisheries-relevant data](#)
- Comox Valley Naturalists: '[Community science for crabs - Tracking larval Dungeness across the Salish Sea](#)'
- University of Quadra: 'Sentinels of Change'
- Tula Quarterly: '[Tracking the Sentinels of Change](#)'
- Hornby Island Natural History Centre: '[Sentinels of Change](#)'

Postseason Gathering

Following our 2024 data collection season, light trap coordinators joined us at the Quadra Centre for Coastal Dialogue, on Quadra Island to debrief and learn from another season together. Attending partners were given an overview of the season and how initial results compare to 2022 and 2023. In an effort to continue improving our network, we discussed challenges, lessons learned, and brainstorming solutions around 3 main topics; equipment monitoring techniques, community engagement, and research priorities. Due to space limitations, we unfortunately could not invite all light trap participants and will gladly provide a summary document from the gathering to those who were not able to attend.



Coordinators at the light trap gathering on Quadra Island, October 2024, photo by Mandy Hart.

School and Community Presentations

In 2024 we presented to several communities, before, during and after the monitoring season. Presentations can be a useful way to engage with communities to encourage participation in the project, and to summarize findings across the network and at specific sites at the end of the season. As this is a collaborative project, time is always budgeted to include discussion and questions from community members and participants. Let us know if you would like us to give a presentation in your community during or after the 2025 season!

Takeaways and Future Years

In our third year of monitoring we had strong community participation with 17 sites returning for a 3rd year, 6 sites returning for a second year and 6 new sites joining the network in 2024. We saw high engagement from our community partners who were dedicated to quality data collection and problem solving through the season. Partners even went above and beyond to engage their communities by involving school groups, delivering community presentations, and communicating about the project on social media to reach a wider audience.

We are now beginning to carry out analyses to identify patterns of larval Dungeness dispersal and abundance and explore some of the factors driving these patterns. In 2026, four years will have elapsed since our first season of data collection, the typical number of years it takes for a Dungeness crab megalopa to reach legal harvest size. This will allow us to compare light trap catch to commercial landings and start investigating the forecasting potential of our dataset. This will be a strong focus next year and in years to come as we continue to collect data. We will continue to work closely with our many partners and with the PCRG to produce a high quality transboundary dataset that will be useful to Dungeness crab managers across the coast.

What's Next?

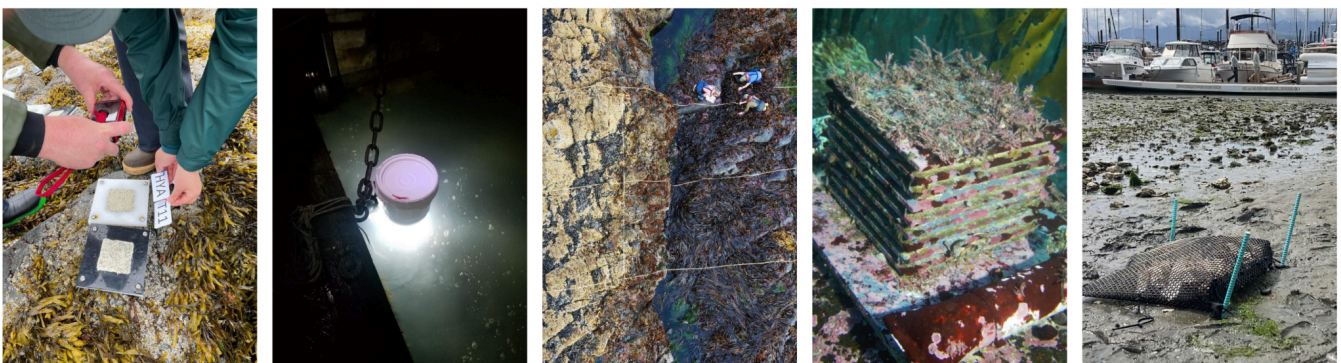
We look forward to many of our 2024 partners returning for a 2nd, 3rd or 4th season with the addition of some new sites in 2025. Next year there will be further experiments with the Marna Wet Lab on Quadra Island to investigate the effects of climate change on early Dungeness life stages and we will continue working with DFO to collect water samples from light trap catches to investigate the potential for the early detection of invasive EGC.

Dungeness crab populations genetics

In 2025, we are launching a population genetics study using larval Dungeness crab caught in the Sentinels of Change light traps. In doing so, we aim to improve our understanding of the genetic connectivity of Dungeness crab populations across the BC coast to gain insight into the extent of larval dispersal and to determine whether there are distinct, self-sustaining populations of Dungeness crab on our coast. Understanding connectivity between crab populations and how it may vary from year to year, is critical in defining the appropriate scale of management for the fishery.

Sentinels Alliance Program

2024 marked a significant step forward in our efforts to implement observation systems designed to study and track biodiversity and change in the Salish Sea. Alongside our community collaborators, several new biodiversity monitoring projects were launched including: Autonomous Reef Monitoring Structures (ARMS), black and white settlement plates, MARINE Biodiversity Surveys, and Oyster Bags. These monitoring systems and surveys gather essential data to test hypotheses about the drivers of biodiversity change and to develop tools that support decision-making and forecasting biodiversity shifts in the Salish Sea. You can read more about specific monitoring projects on our website (<https://sentinels.hakai.org>) and we welcome anyone who has interest and capacity for additional monitoring projects to reach out to us directly.



Sentinels Alliance Program surveys from left to right: black and white tile settlement plates, light traps, MARINE surveys, ARMS, and oyster bags. Photos by Hakai staff.

Acknowledgements

Once again, we extend a huge thank you to our wonderful community partners for the energy, dedication, and passion they bring to the Sentinels light trap network. We are continuously inspired by the care you show for Dungeness crab and the health of our coast. We are grateful to work and learn alongside you - this work could not happen without you!



Photos of community partners submitted on Device Magic throughout the 2024 season!

We are very thankful for our continued collaboration with the Pacific Northwest Crab Research Group. They inspired this work and are a powerful example of what a collaborative network can do. We thank our Hakai Institute colleagues who continue to support this project with their time, expertise, and energy! We are grateful to partner with Fisheries and Oceans Canada to expand this work beyond the Salish Sea to other important areas of the coast.

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